

Another Mother — PostgreSQL Data Schema Outline

Version 1.0 | MVP Phase 1

1. CORE PRINCIPLES

- **Data sovereignty:** All data stored in Australian region (VentralP)
 - **Privacy Act compliance:** Encryption at rest (AES-256) and in transit (TLS 1.3)
 - **Audit trail:** All financial and safety-critical tables have `created_at`, `updated_at`, `version` columns
 - **Soft deletes:** All user-facing tables use `deleted_at` (timestamp) rather than hard deletes
 - **UUID primary keys:** All tables use `uuid` (v4) for security and merge safety
-

2. TABLE SCHEMAS

2.1 USERS & AUTHENTICATION

```
-- Core user table (polymorphic base for families and providers)
CREATE TABLE users (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    email VARCHAR(255) UNIQUE NOT NULL,
    phone VARCHAR(20) UNIQUE NOT NULL,
    password_hash VARCHAR(255) NOT NULL,
    user_type VARCHAR(20) NOT NULL CHECK (user_type IN ('family', 'provider', 'admin')),
    status VARCHAR(20) NOT NULL DEFAULT 'pending' CHECK (status IN ('pending', 'active', 'suspended',
        ↳ 'deactivated')),
    email_verified BOOLEAN DEFAULT FALSE,
    phone_verified BOOLEAN DEFAULT FALSE,
    last_login_at TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    deleted_at TIMESTAMP
);

-- Indexes for auth lookups
CREATE INDEX idx_users_email ON users(email) WHERE deleted_at IS NULL;
CREATE INDEX idx_users_phone ON users(phone) WHERE deleted_at IS NULL;
CREATE INDEX idx_users_type_status ON users(user_type, status) WHERE deleted_at IS NULL;
```

2.2 FAMILY PROFILES

```
CREATE TABLE family_profiles (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,

    -- Identity
    first_name VARCHAR(100) NOT NULL,
    last_name VARCHAR(100) NOT NULL,
    suburb VARCHAR(100) NOT NULL,
```

```

postcode VARCHAR(10) NOT NULL,
state VARCHAR(10) NOT NULL CHECK (state IN ('NSW', 'VIC', 'QLD', 'WA', 'SA', 'TAS', 'ACT', 'NT')),

-- Geolocation (for matching)
latitude DECIMAL(10, 8),
longitude DECIMAL(11, 8),

-- Family composition
children_count INTEGER DEFAULT 0,
children_ages INTEGER[], -- Array of ages: {3, 7, 10}
special_needs_notes TEXT,

-- Preferences
preferred_contact_method VARCHAR(20) DEFAULT 'app' CHECK (preferred_contact_method IN ('app',
↪ 'sms', 'phone')),
emergency_contact_name VARCHAR(100),
emergency_contact_phone VARCHAR(20),

-- Onboarding
onboarding_completed BOOLEAN DEFAULT FALSE,
first_job_posted_at TIMESTAMP,

created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
deleted_at TIMESTAMP
);

CREATE INDEX idx_family_profiles_location ON family_profiles USING GIST (
    ll_to_earth(latitude, longitude)
) WHERE deleted_at IS NULL;
CREATE INDEX idx_family_profiles_suburb ON family_profiles(suburb, postcode) WHERE deleted_at IS NULL;

```

2.3 PROVIDER PROFILES

```

CREATE TABLE provider_profiles (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    user_id UUID NOT NULL REFERENCES users(id) ON DELETE CASCADE,

-- Identity
first_name VARCHAR(100) NOT NULL,
last_name VARCHAR(100) NOT NULL,
display_name VARCHAR(100), -- "Sarah M."
bio TEXT,
photo_url VARCHAR(500),

-- Location
suburb VARCHAR(100) NOT NULL,
postcode VARCHAR(10) NOT NULL,
state VARCHAR(10) NOT NULL,
latitude DECIMAL(10, 8),
longitude DECIMAL(11, 8),

```

```

service_radius_km INTEGER DEFAULT 5 CHECK (service_radius_km BETWEEN 1 AND 50),

-- Experience
years_of_experience INTEGER,
is_mother BOOLEAN DEFAULT TRUE, -- Core brand positioning
own_children_ages INTEGER[],

-- Verification status
verification_status VARCHAR(20) DEFAULT 'pending' CHECK (verification_status IN ('pending',
↳ 'in_review', 'approved', 'rejected', 'expired')),
verified_at TIMESTAMP,
verified_by UUID REFERENCES users(id), -- Admin who approved

-- Performance metrics (for algorithm)
rating DECIMAL(2, 1) DEFAULT 5.0 CHECK (rating >= 1.0 AND rating <= 5.0),
total_reviews INTEGER DEFAULT 0,
completed_jobs INTEGER DEFAULT 0,
completion_rate DECIMAL(5, 2) DEFAULT 100.00, -- Percentage
decline_rate DECIMAL(5, 2) DEFAULT 0.00,

-- Settings
auto_accept_enabled BOOLEAN DEFAULT FALSE,
auto_accept_radius_km INTEGER DEFAULT 5,
max_jobs_per_day INTEGER DEFAULT 3,
minimum_job_value DECIMAL(10, 2) DEFAULT 20.00,

-- Availability status
is_available_now BOOLEAN DEFAULT FALSE,

-- Onboarding
stripe_connect_account_id VARCHAR(255), -- Stripe Connect account
onboarding_completed BOOLEAN DEFAULT FALSE,
went_live_at TIMESTAMP,

created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
deleted_at TIMESTAMP
);

CREATE INDEX idx_provider_profiles_location ON provider_profiles USING GIST (
    ll_to_earth(latitude, longitude)
) WHERE deleted_at IS NULL AND verification_status = 'approved';
CREATE INDEX idx_provider_profiles_verification ON provider_profiles(verification_status, rating) WHERE
↳ deleted_at IS NULL;
CREATE INDEX idx_provider_profiles_algorithm ON provider_profiles(
    verification_status, rating, completion_rate, is_available_now
) WHERE deleted_at IS NULL;

```

2.4 VERIFICATION DOCUMENTS

```
CREATE TABLE verification_documents (
```

```

id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
provider_id UUID NOT NULL REFERENCES provider_profiles(id) ON DELETE CASCADE,

document_type VARCHAR(50) NOT NULL CHECK (document_type IN (
    'wwcc',                -- Working With Children Check
    'police_check',        -- National Police Check
    'cpr_first_aid',       -- CPR/First Aid Certificate
    'photo_id',            -- Driver Licence/Passport
    'public_liability_insurance', -- Insurance certificate
    'other'
)),

-- Document storage
file_url VARCHAR(500) NOT NULL, -- S3/R2 encrypted URL
file_name VARCHAR(255),
file_size_bytes INTEGER,
mime_type VARCHAR(50),

-- Verification metadata
document_number VARCHAR(100), -- WWCC number, etc.
expiry_date DATE,
issuing_authority VARCHAR(100),

-- Review workflow
status VARCHAR(20) DEFAULT 'pending' CHECK (status IN ('pending', 'in_review', 'approved',
    ↪ 'rejected', 'expired')),
reviewed_by UUID REFERENCES users(id),
reviewed_at TIMESTAMPTZ,
review_notes TEXT,

-- Auto-checks
ocr_extracted_data JSONB, -- Extracted text/data from document
verification_check_data JSONB, -- API check results (WWCC registry, etc.)

created_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMPTZ DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_verification_documents_provider ON verification_documents(provider_id, document_type);
CREATE INDEX idx_verification_documents_status ON verification_documents(status) WHERE status IN
    ↪ ('pending', 'in_review');

```

2.5 SERVICE CATEGORIES

```

CREATE TABLE service_categories (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

    name VARCHAR(100) NOT NULL,
    slug VARCHAR(100) UNIQUE NOT NULL,
    description TEXT,
    icon_url VARCHAR(500),

```

```

-- Auto-assign eligibility
requires_wvcc BOOLEAN DEFAULT TRUE,
requires_police_check BOOLEAN DEFAULT TRUE,
min_provider_rating DECIMAL(2, 1) DEFAULT 4.5,
min_completed_jobs INTEGER DEFAULT 5,

-- Pricing guidance
suggested_rate_per_hour DECIMAL(10, 2),

display_order INTEGER DEFAULT 0,
is_active BOOLEAN DEFAULT TRUE,

created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- Seed data
INSERT INTO service_categories (name, slug, description, requires_wvcc, requires_police_check,
↵ suggested_rate_per_hour) VALUES
('School Pickups', 'school-pickups', 'Safe pickup and transport from school', TRUE, TRUE, 35.00),
('Meal Prep', 'meal-prep', 'Home-cooked meals and meal planning', FALSE, FALSE, 30.00),
('Errands & Shopping', 'errands', 'Grocery shopping and local errands', FALSE, FALSE, 28.00),
('Light Household', 'household', 'Tidying, laundry, home reset', FALSE, FALSE, 30.00),
('Short-term Childcare', 'childcare', 'In-home childcare for a few hours', TRUE, TRUE, 32.00),
('Home Reset', 'home-reset', 'Deep tidy and organisational support', FALSE, FALSE, 35.00),
('Laundry', 'laundry', 'Wash, fold, and organise', FALSE, FALSE, 25.00),
('Postpartum Support', 'postpartum', 'Support for new mothers and families', TRUE, TRUE, 40.00),
('Crisis Cover', 'crisis-cover', 'Emergency last-minute support', TRUE, TRUE, 45.00);

```

2.6 PROVIDER SERVICES (Many-to-Many)

```

CREATE TABLE provider_services (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  provider_id UUID NOT NULL REFERENCES provider_profiles(id) ON DELETE CASCADE,
  service_category_id UUID NOT NULL REFERENCES service_categories(id) ON DELETE CASCADE,

  -- Provider-specific settings per service
  is_active BOOLEAN DEFAULT TRUE,
  custom_rate_per_hour DECIMAL(10, 2), -- NULL = use category default
  max_travel_km INTEGER,

  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  UNIQUE(provider_id, service_category_id)
);

```

2.7 AVAILABILITY CALENDAR

```

CREATE TABLE provider_availability (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  provider_id UUID NOT NULL REFERENCES provider_profiles(id) ON DELETE CASCADE,

```

```

-- Time slot
day_of_week INTEGER NOT NULL CHECK (day_of_week BETWEEN 0 AND 6), -- 0=Sunday
start_time TIME NOT NULL,
end_time TIME NOT NULL,

-- Recurring or one-off
is_recurring BOOLEAN DEFAULT TRUE,
specific_date DATE, -- NULL if recurring

-- Status
is_available BOOLEAN DEFAULT TRUE, -- FALSE = blocked out

created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_provider_availability_provider ON provider_availability(provider_id, day_of_week,
↪ start_time) WHERE is_available = TRUE;

```

2.8 JOBS / BOOKINGS

```

CREATE TYPE job_status AS ENUM (
'draft',          -- Family composing request
'posted',         -- Live, seeking provider
'matching',       -- Auto-assign algorithm running
'offered',        -- Offer sent to provider, awaiting response
'accepted',       -- Provider accepted
'confirmed',      -- Payment held, fully confirmed
'en_route',       -- Provider on the way
'in_progress',    -- Job started
'completed',      -- Provider marked complete
'verified_complete', -- Family confirmed completion
'cancelled',      -- Cancelled by either party
'disputed',       -- Under dispute resolution
'refunded',       -- Payment refunded
'expired'         -- No provider found in time
);

CREATE TABLE jobs (
id UUID PRIMARY KEY DEFAULT gen_random_uuid(),

-- Parties
family_id UUID NOT NULL REFERENCES family_profiles(id),
provider_id UUID REFERENCES provider_profiles(id), -- NULL until assigned

-- Service details
service_category_id UUID NOT NULL REFERENCES service_categories(id),
title VARCHAR(255) NOT NULL, -- Auto-generated or family-written
description TEXT, -- Free-text request

-- Location

```

```

address_line_1 VARCHAR(255),
address_line_2 VARCHAR(255),
suburb VARCHAR(100) NOT NULL,
postcode VARCHAR(10) NOT NULL,
latitude DECIMAL(10, 8),
longitude DECIMAL(11, 8),

-- Timing
scheduled_date DATE NOT NULL,
start_time TIME NOT NULL,
end_time TIME,
duration_minutes INTEGER,
is_recurring BOOLEAN DEFAULT FALSE,
recurring_pattern VARCHAR(50), -- "weekly:1,3,5" (Mon,Wed,Fri)
parent_job_id UUID REFERENCES jobs(id), -- For recurring series

-- Status
status job_status DEFAULT 'draft',

-- Matching
matching_algorithm_version VARCHAR(20) DEFAULT '1.0',
matching_score DECIMAL(5, 2), -- Score of assigned provider
assignment_pool VARCHAR(20), -- 'Pool_A', 'Pool_B', 'Pool_C', 'manual'

-- Financial
service_amount DECIMAL(10, 2) NOT NULL, -- Provider earnings
platform_fee DECIMAL(10, 2) NOT NULL, -- 15%
total_amount DECIMAL(10, 2) NOT NULL, -- service + fee
stripe_payment_intent_id VARCHAR(255),
stripe_transfer_id VARCHAR(255), -- For provider payout

-- Cancellation
cancelled_by UUID REFERENCES users(id),
cancelled_at TIMESTAMP,
cancellation_reason TEXT,
cancellation_fee_applied DECIMAL(10, 2) DEFAULT 0.00,

-- Completion
completed_at TIMESTAMP,
verified_complete_at TIMESTAMP,

-- Metadata
special_instructions TEXT,
children_present JSONB, -- [{"name": "Emma", "age": 5, "notes": "..."}]

created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_jobs_family ON jobs(family_id, status) WHERE status NOT IN ('completed', 'cancelled',
↵ 'refunded');

```

```
CREATE INDEX idx_jobs_provider ON jobs(provider_id, status) WHERE status NOT IN ('completed',
↪ 'cancelled', 'refunded');
CREATE INDEX idx_jobs_matching ON jobs(status, service_category_id, scheduled_date) WHERE status IN
↪ ('posted', 'matching');
CREATE INDEX idx_jobs_location ON jobs USING GIST (ll_to_earth(latitude, longitude)) WHERE status IN
↪ ('posted', 'matching');
CREATE INDEX idx_jobs_recurring ON jobs(parent_job_id) WHERE is_recurring = TRUE;
```

2.9 JOB OFFERS (Auto-Assign Cascade)

```
CREATE TABLE job_offers (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  job_id UUID NOT NULL REFERENCES jobs(id) ON DELETE CASCADE,
  provider_id UUID NOT NULL REFERENCES provider_profiles(id),

  -- Offer state
  status VARCHAR(20) DEFAULT 'pending' CHECK (status IN ('pending', 'accepted', 'declined', 'expired',
↪ 'superseded')),

  -- Scoring
  matching_score DECIMAL(5, 2) NOT NULL,
  matching_factors JSONB, -- {"proximity": 85, "availability": 100, ...}

  -- Response
  responded_at TIMESTAMP,
  decline_reason VARCHAR(50), -- 'too_far', 'time_conflict', 'comfort_zone', 'personal', 'other'
  decline_note TEXT,

  -- Notifications
  push_sent_at TIMESTAMP,
  sms_sent_at TIMESTAMP,
  push_acknowledged_at TIMESTAMP,

  -- Cascade tracking
  offer_sequence INTEGER NOT NULL DEFAULT 1, -- 1st, 2nd, 3rd offer

  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  expires_at TIMESTAMP NOT NULL -- Usually created_at + 3 minutes
);

CREATE INDEX idx_job_offers_job ON job_offers(job_id, offer_sequence);
CREATE INDEX idx_job_offers_provider ON job_offers(provider_id, status) WHERE status = 'pending';
CREATE INDEX idx_job_offers_expiry ON job_offers(expires_at) WHERE status = 'pending';
```

2.10 SAVED PROVIDERS (“My Mothers”)

```
CREATE TABLE saved_providers (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  family_id UUID NOT NULL REFERENCES family_profiles(id) ON DELETE CASCADE,
  provider_id UUID NOT NULL REFERENCES provider_profiles(id) ON DELETE CASCADE,
```



```

-- Relationship metadata
nickname VARCHAR(100), -- "Our Sarah"
total_jobs_together INTEGER DEFAULT 0,
last_job_date DATE,

-- Preference
is_preferred_regular BOOLEAN DEFAULT FALSE, -- Prioritise for auto-assign

created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
UNIQUE(family_id, provider_id)
);

CREATE INDEX idx_saved_providers_family ON saved_providers(family_id, is_preferred_regular) WHERE
↪ is_preferred_regular = TRUE;

```

2.11 MESSAGES

```

CREATE TABLE messages (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  job_id UUID NOT NULL REFERENCES jobs(id) ON DELETE CASCADE,
  sender_id UUID NOT NULL REFERENCES users(id),
  receiver_id UUID NOT NULL REFERENCES users(id),

  content TEXT NOT NULL,
  message_type VARCHAR(20) DEFAULT 'text' CHECK (message_type IN ('text', 'system', 'image',
↪ 'location')),

-- Media
attachment_url VARCHAR(500),

-- Status
read_at TIMESTAMP,
delivered_at TIMESTAMP,

created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_messages_job ON messages(job_id, created_at);
CREATE INDEX idx_messages_receiver ON messages(receiver_id, read_at) WHERE read_at IS NULL;

```

2.12 REVIEWS

```

CREATE TABLE reviews (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  job_id UUID NOT NULL REFERENCES jobs(id) ON DELETE CASCADE,

-- Direction
reviewer_id UUID NOT NULL REFERENCES users(id), -- Who wrote it
reviewee_id UUID NOT NULL REFERENCES users(id), -- Who it's about
reviewer_type VARCHAR(20) NOT NULL CHECK (reviewer_type IN ('family', 'provider')),

```

```

-- Content
rating INTEGER NOT NULL CHECK (rating BETWEEN 1 AND 5),
text_content TEXT,

-- Categories
was_punctual BOOLEAN,
was_communicative BOOLEAN,
was_caring BOOLEAN,
would_rebook BOOLEAN,

-- Moderation
status VARCHAR(20) DEFAULT 'published' CHECK (status IN ('published', 'flagged', 'hidden',
↳ 'under_review')),
flagged_reason TEXT,

-- Response
response_text TEXT,
responded_at TIMESTAMP,

created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
UNIQUE(job_id, reviewer_id) -- One review per direction per job
);

CREATE INDEX idx_reviews_reviewee ON reviews(reviewee_id, status, created_at) WHERE status =
↳ 'published';

```

2.13 PAYMENTS & TRANSACTIONS

```

CREATE TABLE transactions (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  job_id UUID NOT NULL REFERENCES jobs(id),

  -- Transaction details
  transaction_type VARCHAR(50) NOT NULL CHECK (transaction_type IN (
    'payment_hold',      -- Family charged, funds held
    'platform_fee',      -- 15% retained
    'provider_payout',   -- 85% to provider
    'refund',            -- Full or partial refund
    'cancellation_fee',  -- Fee charged on late cancellation
    'instant_payout_fee' -- Stripe instant payout fee
  )),

  -- Parties
  from_party UUID REFERENCES users(id), -- NULL if from Stripe/customer
  to_party UUID REFERENCES users(id),   -- NULL if to Stripe/platform

  -- Amounts
  amount DECIMAL(10, 2) NOT NULL,
  currency VARCHAR(3) DEFAULT 'AUD',

  -- Stripe references

```

```

stripe_payment_intent_id VARCHAR(255),
stripe_transfer_id VARCHAR(255),
stripe_charge_id VARCHAR(255),
stripe_balance_transaction_id VARCHAR(255),

-- Status
status VARCHAR(20) DEFAULT 'pending' CHECK (status IN ('pending', 'processing', 'completed',
↪ 'failed', 'reversed')),

-- Metadata
description TEXT,
failure_reason TEXT,

created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
processed_at TIMESTAMP,

UNIQUE(job_id, transaction_type, status) WHERE status = 'completed'
);

CREATE INDEX idx_transactions_job ON transactions(job_id, transaction_type);
CREATE INDEX idx_transactions_stripe ON transactions(stripe_payment_intent_id) WHERE
↪ stripe_payment_intent_id IS NOT NULL;

```

2.14 PROVIDER EARNINGS & PAYOUTS

```

CREATE TABLE provider_earnings (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  provider_id UUID NOT NULL REFERENCES provider_profiles(id) ON DELETE CASCADE,

  -- Balance tracking
  available_balance DECIMAL(10, 2) DEFAULT 0.00,
  pending_balance DECIMAL(10, 2) DEFAULT 0.00, -- Jobs completed, not yet released
  lifetime_earnings DECIMAL(10, 2) DEFAULT 0.00,

  -- Payout settings
  payout_method VARCHAR(20) DEFAULT 'auto_weekly' CHECK (payout_method IN ('auto_weekly', 'manual',
↪ 'instant')),
  auto_payout_threshold DECIMAL(10, 2) DEFAULT 50.00,
  last_payout_at TIMESTAMP,

  updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE payout_history (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  provider_id UUID NOT NULL REFERENCES provider_profiles(id) ON DELETE CASCADE,

  amount DECIMAL(10, 2) NOT NULL,
  payout_method VARCHAR(20) NOT NULL, -- 'auto_weekly', 'manual', 'instant'
  stripe_payout_id VARCHAR(255),

```

```

status VARCHAR(20) DEFAULT 'pending' CHECK (status IN ('pending', 'in_transit', 'paid', 'failed',
↳ 'canceled')),

initiated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
completed_at TIMESTAMP,
failure_reason TEXT,

-- Associated transactions
transaction_ids UUID[] -- References transactions table
);

```

2.15 ADMIN & AUDIT

```

CREATE TABLE admin_actions (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  admin_id UUID NOT NULL REFERENCES users(id),

  action_type VARCHAR(50) NOT NULL, -- 'verify_provider', 'suspend_user', 'resolve_dispute', etc.
  target_type VARCHAR(50) NOT NULL, -- 'user', 'provider', 'job', 'document'
  target_id UUID NOT NULL,

  action_data JSONB, -- Before/after state, reason, etc.
  notes TEXT,

  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE incidents (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  reporter_id UUID NOT NULL REFERENCES users(id),

  incident_type VARCHAR(50) NOT NULL CHECK (incident_type IN (
    'safety_concern',
    'no_show',
    'late_arrival',
    'inappropriate_behavior',
    'payment_dispute',
    'property_damage',
    'other'
  )),

  severity VARCHAR(20) DEFAULT 'medium' CHECK (severity IN ('low', 'medium', 'high', 'critical')),

  -- Related entities
  job_id UUID REFERENCES jobs(id),
  reported_user_id UUID REFERENCES users(id),

  description TEXT NOT NULL,
  evidence_urls TEXT[], -- S3 URLs

  -- Resolution

```

```

status VARCHAR(20) DEFAULT 'open' CHECK (status IN ('open', 'investigating', 'resolved',
↪ 'escalated', 'closed')),
resolved_by UUID REFERENCES users(id),
resolution_notes TEXT,
resolution_action VARCHAR(50), -- 'warning', 'suspension', 'ban', 'refund', 'none'

created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
resolved_at TIMESTAMP
);

CREATE INDEX idx_incidents_status ON incidents(status, severity) WHERE status = 'open';

```

3. REDIS DATA STRUCTURES (For Auto-Assign Engine)

```

# Real-time provider availability index
# Key: geo:providers:available
# Type: Redis Geo (GEORADIUS)
# Value: provider_id with lat/lon
# TTL: 300s (refreshed every 5 min by cron)

# Active job matching queue
# Key: jobs:matching:{job_id}
# Type: Redis Sorted Set (ZSET)
# Score: matching_score
# Member: provider_id
# TTL: Until job is assigned or expires

# Provider offer state
# Key: offer:{job_id}:{provider_id}
# Type: Redis Hash
# Fields: status, expires_at, offer_sequence
# TTL: 3 minutes (offer window)

# Rate limiting
# Key: rate:provider:{provider_id}:offers
# Type: Redis String (INCR with EXPIRE)
# Value: count of offers in last hour
# TTL: 3600s

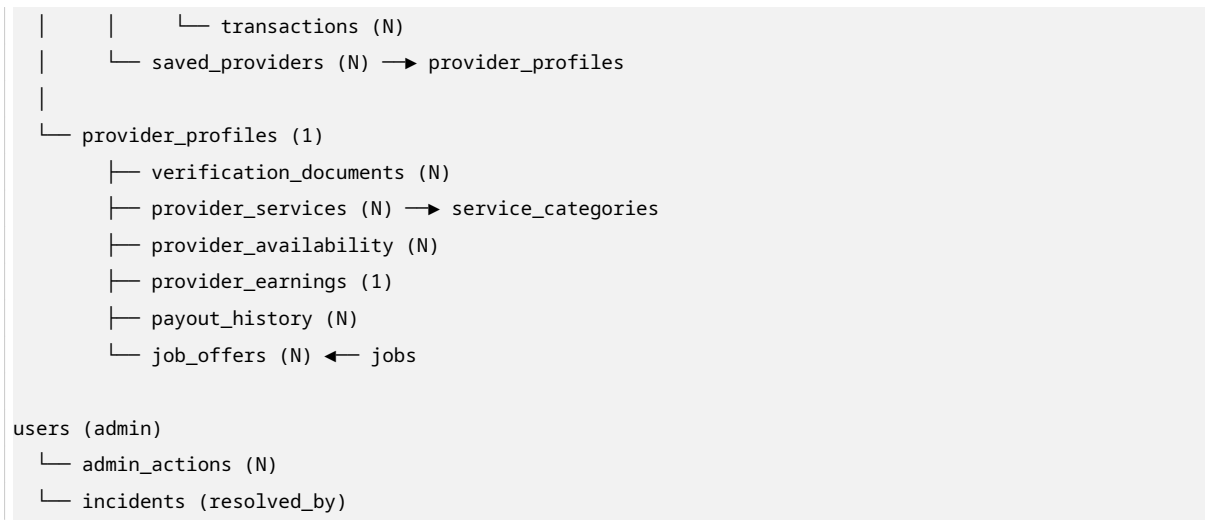
```

4. KEY RELATIONSHIPS (ER Diagram Summary)

```

users (1)
├─ family_profiles (1)
│   └─ jobs (N)
│       └─ job_offers (N)
│           └─ messages (N)
│               └─ reviews (N)

```



5. PERFORMANCE OPTIMISATIONS

Query Pattern	Optimisation
Find providers near location	GIST index on <code>ll_to_earth(lat, lon)</code> + <code>geo:providers:available</code> Redis Geo
Active jobs for family/provider	Partial index excluding completed/cancelled
Pending offers for provider	Composite index on <code>(provider_id, status)</code>
Unread messages	Index on <code>(receiver_id, read_at)</code> with <code>read_at IS NULL</code>
Published reviews	Partial index on <code>status = 'published'</code>
Algorithm scoring	Materialised view refreshed every 60s, Redis pre-computed scores

6. DATA RETENTION & PRIVACY

Data Type	Retention	Action
Messages	2 years post-job	Soft delete, archive to cold storage
Verification documents	7 years (legal)	Encrypted S3, access logged
Job history	7 years	Anonymise after 2 years (remove names, keep metrics)
Failed login attempts	90 days	Auto-purge
Session tokens	Until expiry	Redis TTL
PII (address, phone)	Until account deletion	Crypto-shredded on deletion request

7. MIGRATION STRATEGY

1. **Baseline:** `001_initial_schema.sql` — All tables above
 2. **Seed:** `002_seed_data.sql` — Service categories, admin user
 3. **Functions:** `003_matching_functions.sql` — PostgreSQL functions for algorithm scoring
 4. **Triggers:** `004_audit_triggers.sql` — Auto-update `updated_at`, log changes
 5. **Indexes:** `005_performance_indexes.sql` — All indexes listed above
 6. **Views:** `006_analytics_views.sql` — Pre-built views for admin dashboard
-

8. NEXT SCHEMA ADDITIONS (Post-MVP)

- `subscription_plans` — For priority booking tiers
- `provider_scheduling_blocks` — Advanced calendar with buffer times
- `job_templates` — Saved recurring job patterns
- `referral_codes` — Growth tracking
- `notification_preferences` — Granular opt-in/out settings
- `chat_rooms` — Group/community chat (“village” feature)